Supplementary Material

Report-Supervised Distillation for Multi-Label Knee MRI Classification: An Evaluation Under Limited Expert Annotation

S1. Single-change switches inside the fused ensembles (exploratory)

Table S1 lists changes, other than adding students, that were tested as the single change against a parent on the development partition. All rows were displayed as 0.958 unless noted. The effect is the direction of the server's unrounded ordering of the two submissions: ordinal, not a numerical difference, from comparisons that share parents and are not independent. It should be read as exploratory and is not used for any conclusion in the main text.

Table S1: Single-change switches on 0.958-level parents (development partition, unrounded ordering).

Change	Pairs	Direction (variant vs parent)
Add S6fall-B at 1/3 rank weight	1	above
BK: Baker's-cyst stacking model of R6 disabled	2	above in 2/2
FK: effusion stacking model of R6 disabled	2	above in 2/2
Published per-finding weights V2, with students present	3	above in 3/3
Published per-finding weights V2, without students	1	below
Student weight $w = 0.3$ instead of 0.2	2	below in 2/2
Left/right mirror test-time augmentation of the students	2	below in 2/2
Low-weight published 2.5D reader ("Part B") appended	2	below in 2/2
MaViT model appended at weight 0.1	1	below (displayed 0.957)

S2. Inference safeguards

A submission that fails or exceeds the 9-h limit yields no score, so the student stage was written to fall back to the parent's predictions. Algorithm S1 is the procedure used by the final system; it changes nothing in a run without failures.

Algorithm S1 Rank fusion with fall-back to the parent predictions

Input: parent predictions P (file `sub`); student checkpoints with hashes $\{(\theta_m, h_m)\}_{m=1}^{M}$; time limit $T_{\max}$; elapsed time T_0; weight w

Output: final predictions F

$P_0 \leftarrow$ byte-exact copy of `sub`
verify $\mathrm{SHA256}(\theta_m) = h_m$ for all m; on mismatch **return** P_0
run the students with timeout $T_{\max} - T_0 - 20\,\mathrm{min}$; on error or timeout **return** P_0
$\bar{S} \leftarrow \frac{1}{M}\sum_m \sigma(f_{\theta_m}(X))$
for $c = 1, \ldots, 12$ **do**
    $\rho^P \leftarrow r(P_{\cdot c})$, $\rho^S \leftarrow r(\bar{S}_{\cdot c})$
    $\mathcal{U} \leftarrow \{i : \bar{S}_{ic}$ missing$\}$; fit $\rho^S \approx \alpha + \beta\rho^P$ on $i \notin \mathcal{U}$; impute ρ_i^S for $i \in \mathcal{U}$
    $F_{\cdot c} \leftarrow (1-w)\rho^P + w\rho^S$
end for
if F fails the schema, row-count or finiteness checks **then return** P_0
end if
return F

[TODO: Report the number of studies imputed (line 7) and the number of fall-backs (lines 2, 3, 10) in the final runs, from the inference logs.]

S3. CLAIM 2024 reporting checklist

Items follow the topics of CLAIM 2024 (Tejani et al., *Radiol. Artif. Intell.* 6(4):e240300, 2024). "Unavailable" means the information was not released by the data provider and is not inferred. [TODO: Check every row against the official item list (`https://pubs.rsna.org/page/ai/claim`) and add item numbers before submission.]

Topic	Status	Where / note
Title, abstract	Yes	Title; abstract states the cohort, partitions and that development results are exploratory.
Background, objectives, hypothesis	Yes	Section I (central question and hypothesis).
Study design (retrospective, exploratory)	Yes	Sections III-E, V; development vs final-test partitions.
Data sources	Partial	Section III-A (international, multilingual); institutions and acquisition period unavailable.
Eligibility criteria, cohort flow	Unavailable	Not released by the data provider.
Demographics, scanners, field strength	Unavailable	Not released.
Patient-level identifiers, repeat examinations	Unavailable	Not released; report grouping only (Section III-A).
Data preprocessing	Yes	Section IV-B, Algorithm 2 (including orientation and percentile limitations).
Missing data	Partial	Missing series slots are masked (Section IV-C); decoding errors 0; per-slot missing counts [TODO: add].
Reference standard: definitions	Yes	Table 1 (organisers' rubric; borderline findings negative).
Reference standard: annotators, process	Yes	Section III-A: two subspecialty MSK radiologists, third adjudicates (organiser overview); same process for the test set (organiser reply, discussion 733491).
Selection of the 58 reference studies	Unavailable	Enriched (Table 2, Fig. 5); selection procedure not released.
Use of reference studies during development	Yes	Section III-B (diagnostic, adoption rules, teacher selection).
Data partitions	Yes	Section III-E, Fig. 7; folds in Algorithm 1; release includes fold assignments.
Model description	Yes	Section IV-C, Fig. 8, Algorithm 3, Table 4; reused code identified.
Training details, initialisation	Yes	Section IV-D, Table 4, Algorithm 4.
Teacher / external data provenance	Partial	Table 5; some entries unverified.
Model selection	Yes	Section V (prospectively logged rules); EMA of last epoch.
Ensembling	Yes	Section IV-F, Eq. 5, Table 9, Supplementary S1–S2.
Performance metrics	Yes	Macro ROC AUC; per-finding AUC on report labels (Table 11); label analysis (Section III-D, Figs. 3–6).
Statistical methods, uncertainty	Partial	Bootstrap on $\mathcal{G}_{58}$ (Section V); test-set intervals impossible (labels hidden).
Robustness, subgroup analysis	No	Not performed (Section VIII).
External testing	No	Not performed (Section VIII).

Topic	Status	Where / note
Calibration, operating points	No	Not performed; discussed in Section VII.
Failure analysis	No	Not performed; [TODO: add examples if a qualified reader can review them].
Limitations	Yes	Section VIII.
Implications for practice	Yes	Section VII (clinical considerations).
Data and code availability	Partial	Code and logs released; checkpoints [TODO: location and hashes].
Funding, conflicts of interest	[TODO:]	Front matter.
Ethics approval	[TODO:]	Section VII.